

Chemical Mixing Guide

Lime · PACl Coagulant · Flocculant · SUMAS Water Treatment, Kamloops BC
Created by Oliver Peters · June 2026

1. Lime

1 bag (50 lbs) per 200L of water



TEXLIME High Calcium
Hydrated Lime — confirmed
on-site bag

1. Add 200L of clean water to the mixing tote.
2. Add 1 bag (50 lbs) of lime.
3. Turn on the air mixer (open the red valve) — let it blend.
4. **Keep the mixer running the whole time you're pumping it into the system.** If the blade stops, the lime settles and hardens fast, and the dose won't work.

Critical: this is not "mix once and walk away." The blade has to stay on for as long as you're drawing lime out — mixing and pumping happen at the same time.

2. PACl Coagulant

1 bag (25kg dry powder) per 100L of clean water



Poly Aluminum Chloride, 25kg
bag — confirmed on-site bag

1. Add 100L of clean water to the mixing tote (not the water being treated).
2. Add 1 bag (25kg) of dry PACl powder.
3. Mix until dissolved.
4. **Let it sit 3 hours before using** — don't dose straight from a freshly mixed batch.
5. Meter the resulting solution into the process with the dosing pump.

Plan ahead: mix a new batch before you run out mid-shift — the 3-hour rest means you can't make more on demand.

3. Flocculant

1 kg polymer per 1000L water (or 500g per 500L batch)



Flopham water-soluble polymer — confirmed product label

1. Add the **first half** of the water to the tote.
2. Turn on the mixer.
3. Pour in the polymer **while** adding the second half of the water at the same time — this stops it from clumping.
4. Keep mixing until fully dissolved.
5. **Let it sit 24 hours before use.**

Never dump dry polymer straight into still water — it forms lumps that won't dissolve. Always pour it in while water is moving/mixing.

4. The Mixer (Air-Driven Tote Mixer)



Mixer clamped on top of the tote



Close-up: red air valve, shaft & propeller

1. Connect the air hose to the compressed air supply (quick-connect fitting).
2. Open the **red-handled valve** — air spins the motor, which turns the propeller inside the tote.
3. Open it gradually for flocculant — full-blast air can tear the polymer apart instead of dissolving it. Lime and PACl can run full open.
4. Close the red valve to stop.

Don't confuse it with the nearby cross-handled valve — that one's the tote's drain/outlet, not the mixer.